| Doc Number: 161310G00000-00 | Doc Version: 008 | Date: 02-Nov-2026 |
|---|---|---|

# Super Mushshak Automatic Flight Test

Prepar3D v5 | Rahwali Airfield (OPRW) | Automatic Testing Mode

**Super Mushshak Logger - ver. 1.0**

Super Mushshak Airborne Automatic Testing Suite

Flight Test File: Super_Mushak_auto_takeoff_data_2026-11-02_12-13-52.csv

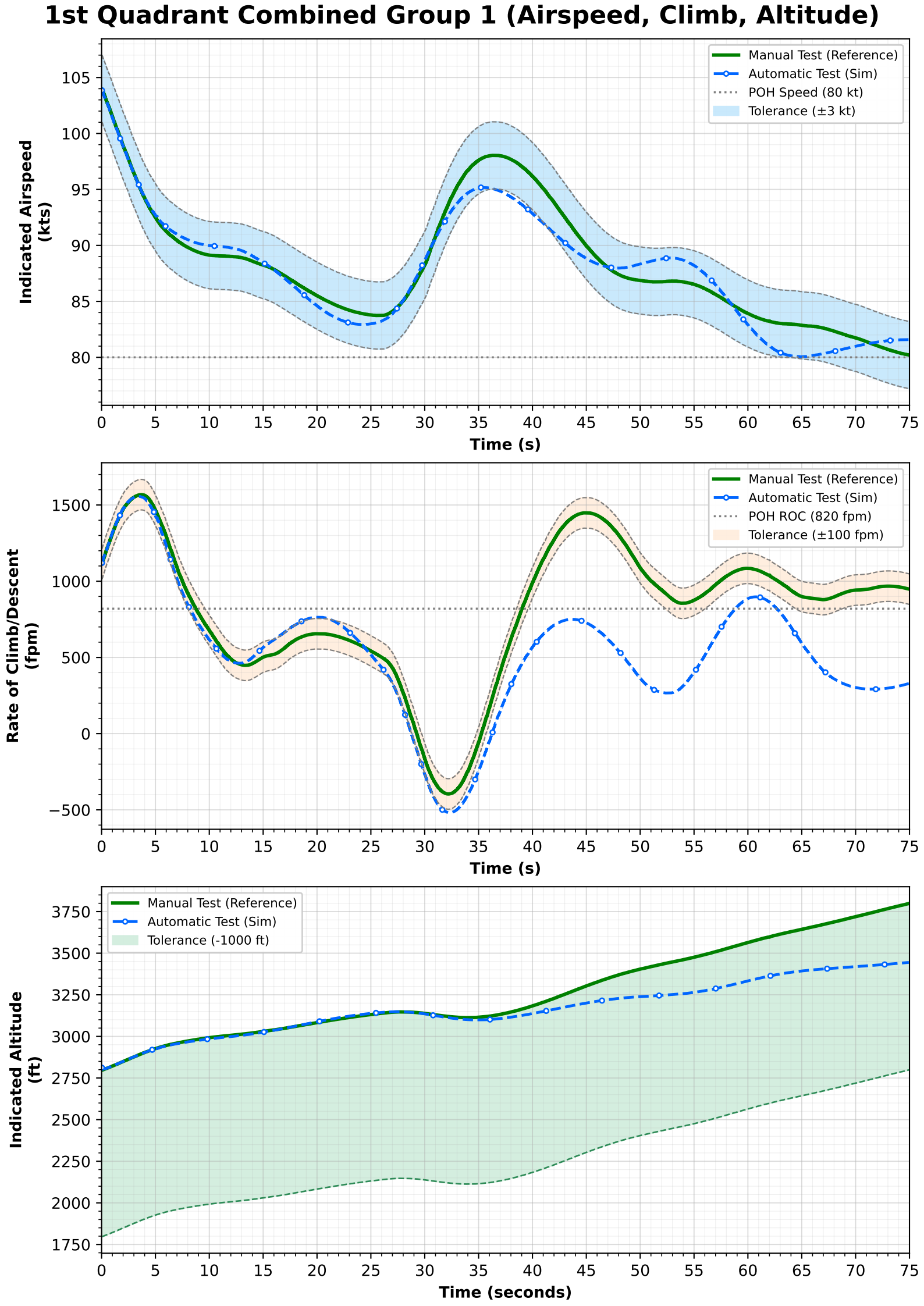
**ONLINE, AUTOMATIC**

Test Date: 02-Nov-2026  Test Time: 12:13:52

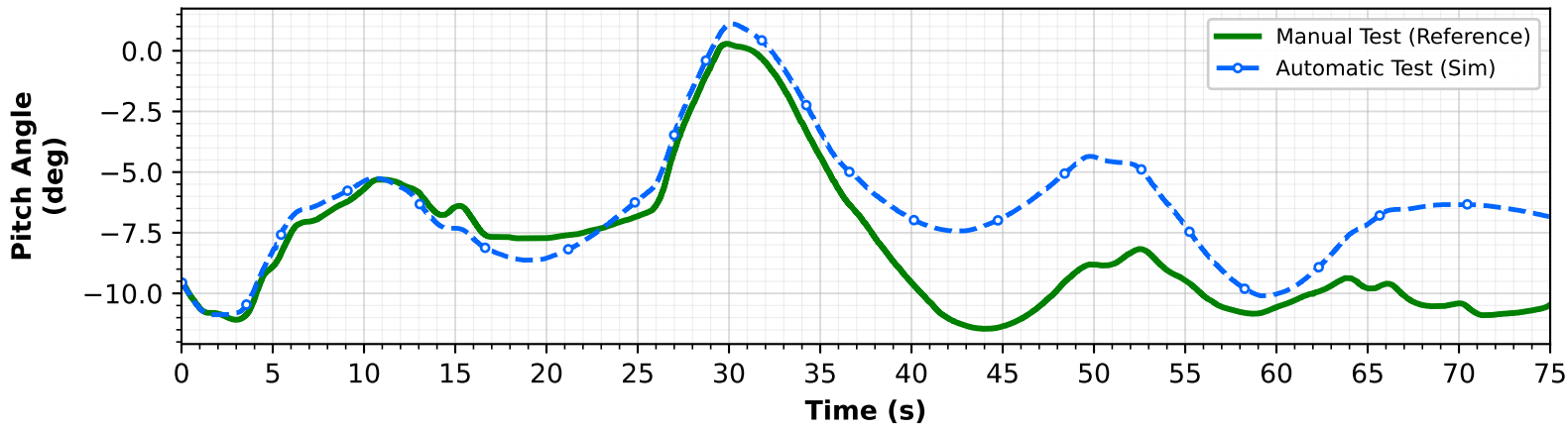
## Initial Test Conditions

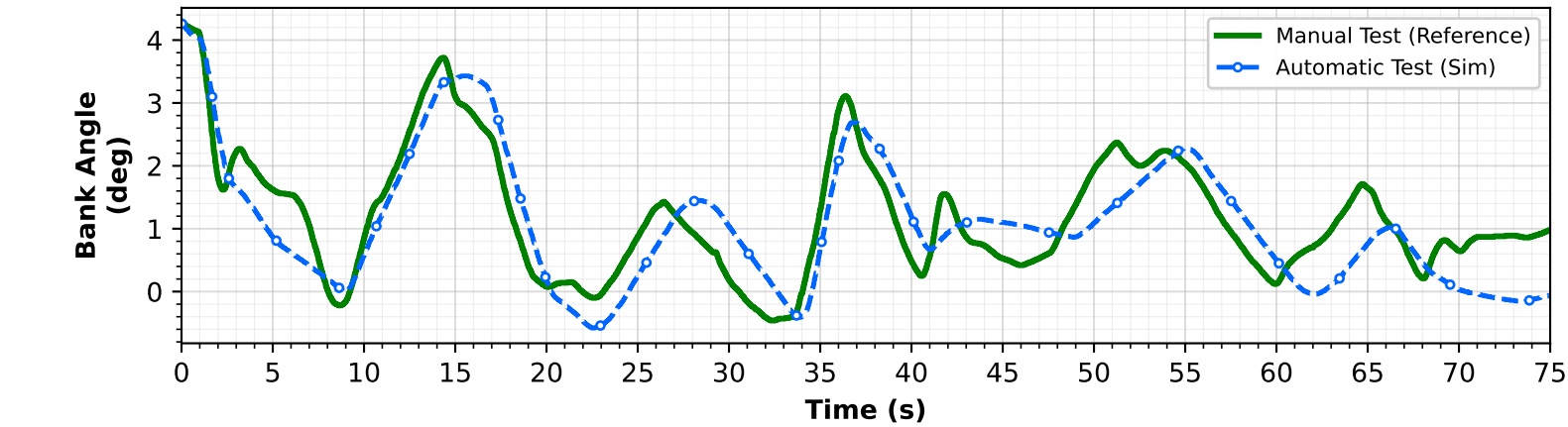
| Validation | Sim | |
|---|---|---|
| ***** Weight and Balance ***** | | |
| 846.2 | 846.2 | Gross Wt (kg) |
| 107.1 | 107.1 | Total Fuel Quantity (kg) |
| 0.00 | 0.00 | CG Pos in X Axis (mm) |
| 0.00 | 0.00 | CG Pos in Y Axis (mm) |
| ***** Control Parameters ***** | | |
| -4.60 | -4.51 | Pitch Control Position (raw) |
| -0.28 | -0.52 | Roll Control Position (raw) |
| 0.00 | 0.00 | Rudder Pedal Position (raw) |
| 81.4 | 81.4 | Throttle Position (%) |
| -0.06 | -0.03 | Pitch Force: Applied/Virtual (N) |
| -0.00 | -0.00 | Roll Force: Applied/Virtual (N) |
| 0.00 | -0.00 | Yaw Force: Applied/Virtual (N) |
| ***** Environment Parameters ***** | | |
| 1013 | 1013 | Baro Press (hPa) |
| 11.0 | 11.0 | OAT (deg C) |
| 325.6 | 325.5 | Wind Dir (mag,deg) |
| 7.60 | 7.60 | Wind Spd (kt) |
| ***** Engine Parameters (Piston Engine) ***** | | |
| 2700 | 2692 | Engine Speed (RPM) |
| 81.4 | 81.4 | Throttle Position (%) |
| 0.00 | 0.00 | Fuel Flow (pph) |
| 535.7 | 533.4 | Exhaust Gas Temp (deg C) |

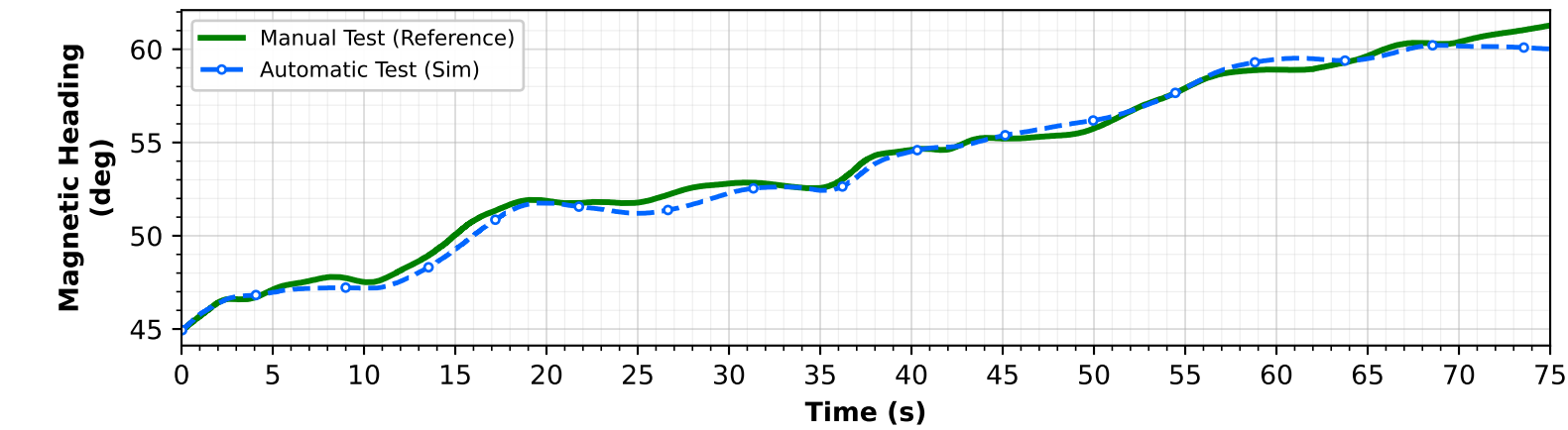
| Validation | Sim | |
|---|---|---|
| ***** Flight Parameters ***** | | |
| 104.0 | 103.8 | Indicated Airspeed (kt) |
| 106.2 | 106.2 | Ground Speed (kt) |
| 0.17 | 0.17 | Mach Number (-) |
| 2800 | 2801 | Pressure Alt (ft) |
| 2055 | 2056 | Radar Alt (ft) |
| 750.0 | 750.0 | Height of Terrain (ft) |
| 1116 | 1120 | Vertical Velocity (ft/min) |
| -9.54 | -9.56 | Pitch Angle (deg) |
| 4.27 | 4.26 | Bank Angle (deg) |
| 44.9 | 44.9 | Heading (deg) |
| 1.77 | 1.75 | Aircraft Angle of Attack (deg) |
| 0.18 | 0.40 | Sideslip Angle (deg) |
| 48.3 | 50.5 | Track Angle (deg) |
| 0.31 | 0.34 | Roll Rate (deg/s) |
| -1.37 | -1.38 | Pitch Rate (deg/s) |
| 0.72 | 0.73 | Yaw Rate (deg/s) |
| 0.02 | -0.08 | X Body Acceleration (g) |
| -0.24 | -0.46 | Y Body Acceleration (g) |
| -4.11 | -4.14 | Z Body Acceleration (g) |

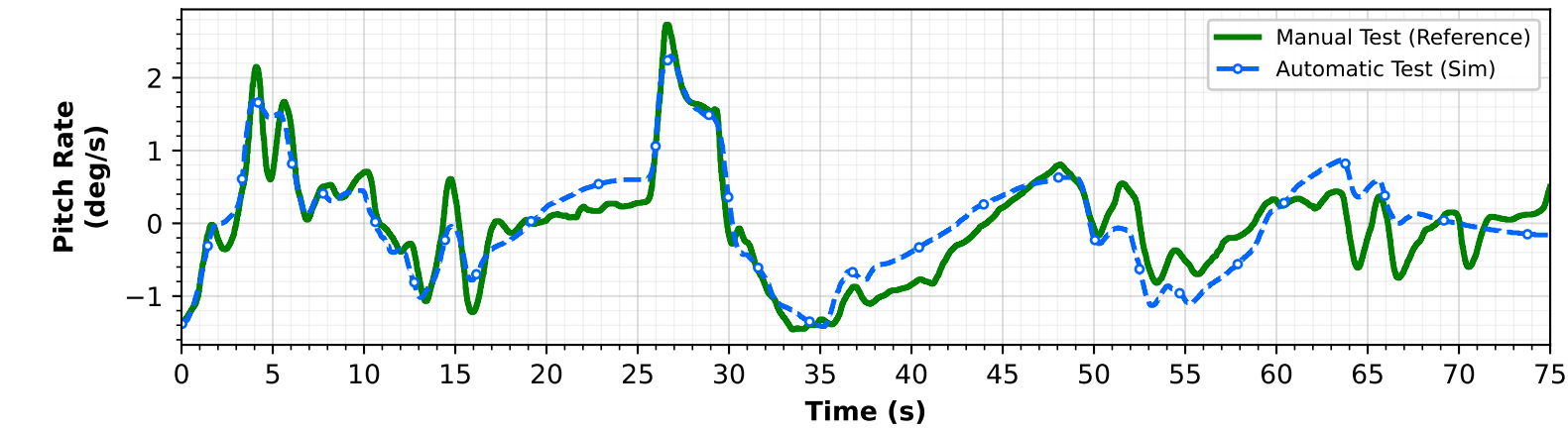

1st Quadrant Combined Group 1 (Airspeed, Climb, Altitude)
Indicated Airspeed (kts)
Manual Test (Reference)
Automatic Test (Sim)
POH Speed (80 kt)
Tolerance (±3 kt)
Time (s)
Rate of Climb/Descent (fpm)
Manual Test (Reference)
Automatic Test (Sim)
POH ROC (820 fpm)
Tolerance (±100 fpm)
Time (s)
Indicated Altitude (ft)
Manual Test (Reference)
Automatic Test (Sim)
Tolerance (-1000 ft)
Time (seconds)

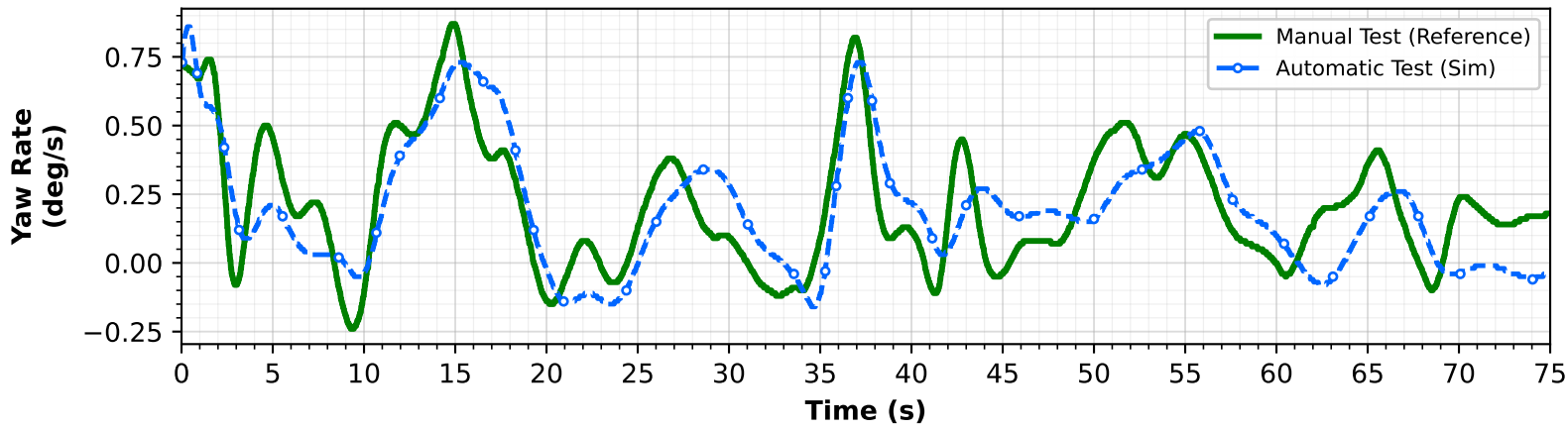
# 1st Quadrant Combined Group 2 (Pitch, Bank, Yaw, Heading)



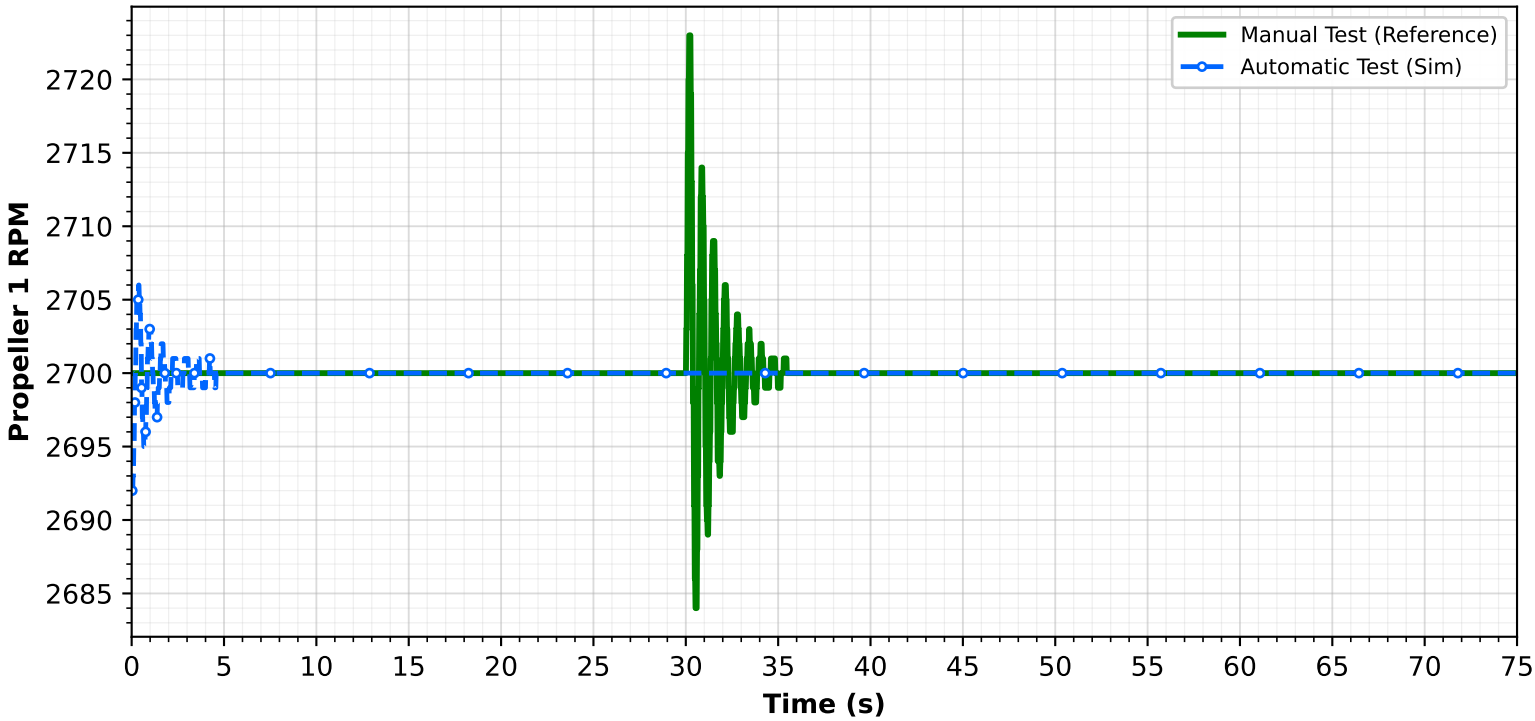



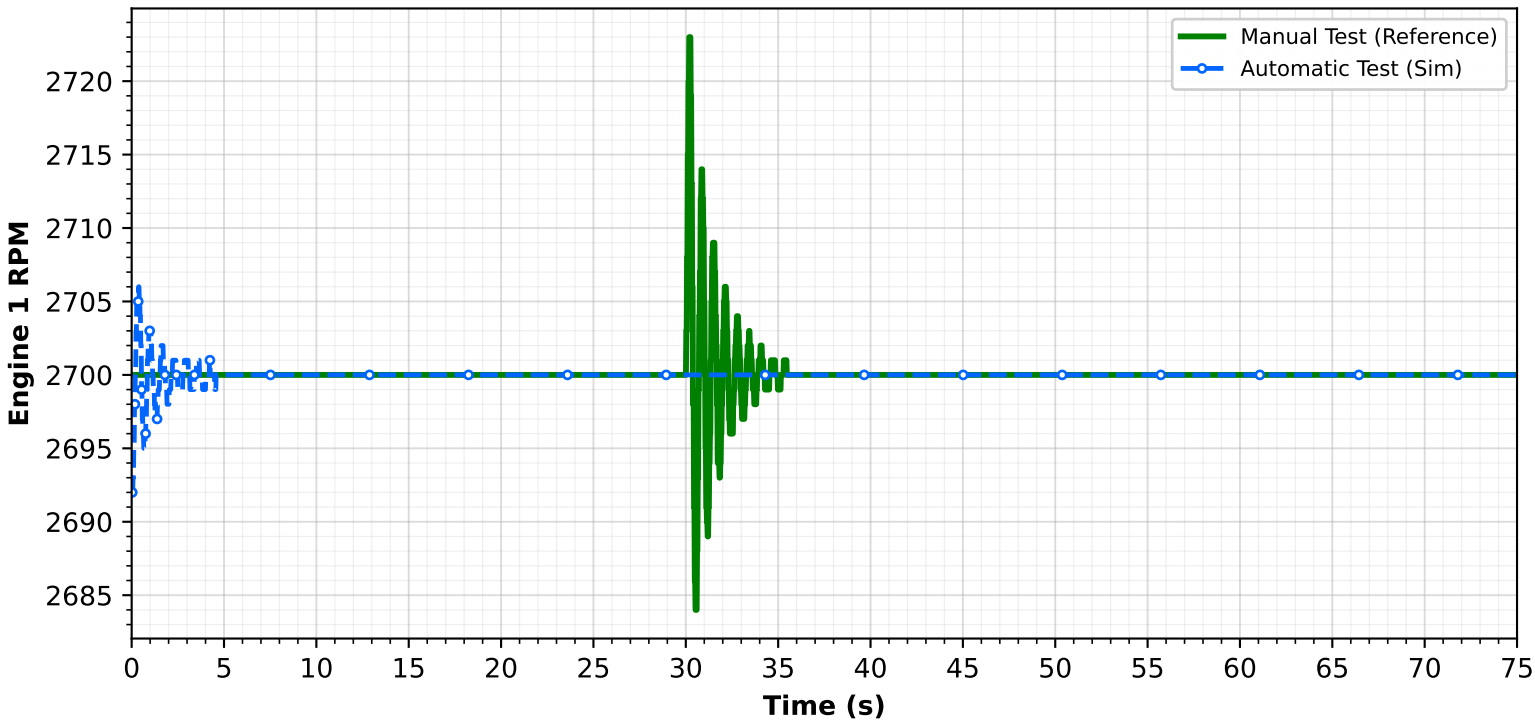



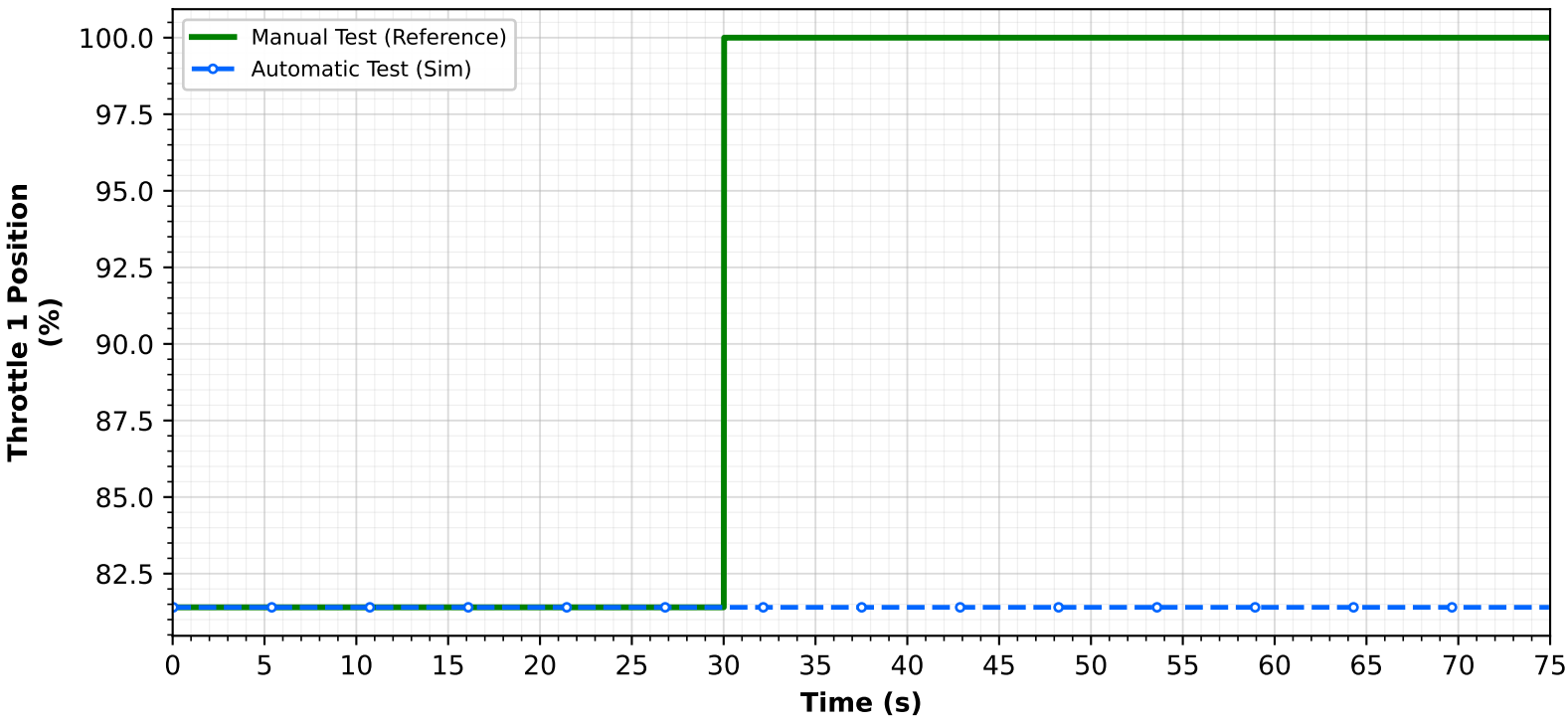



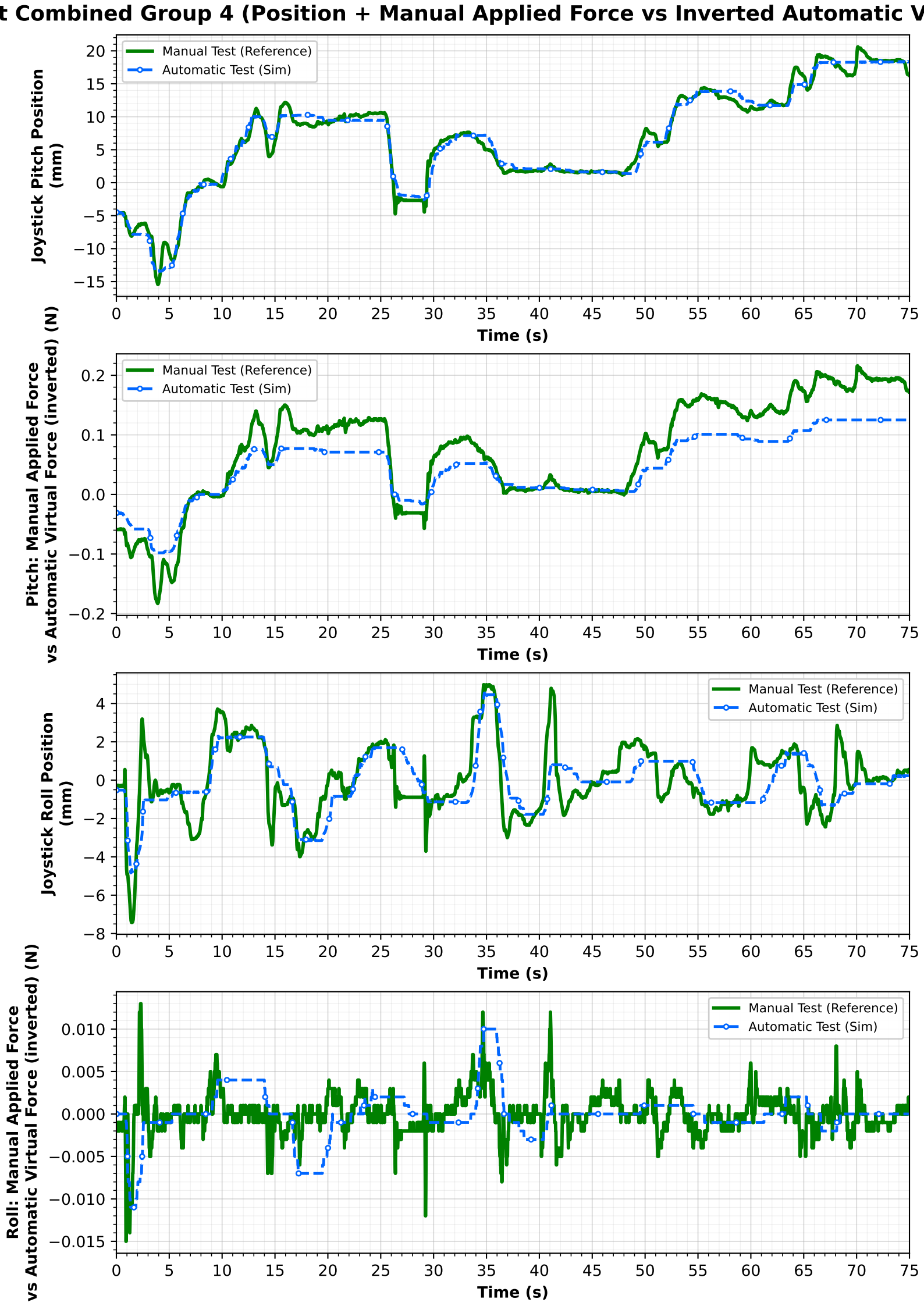



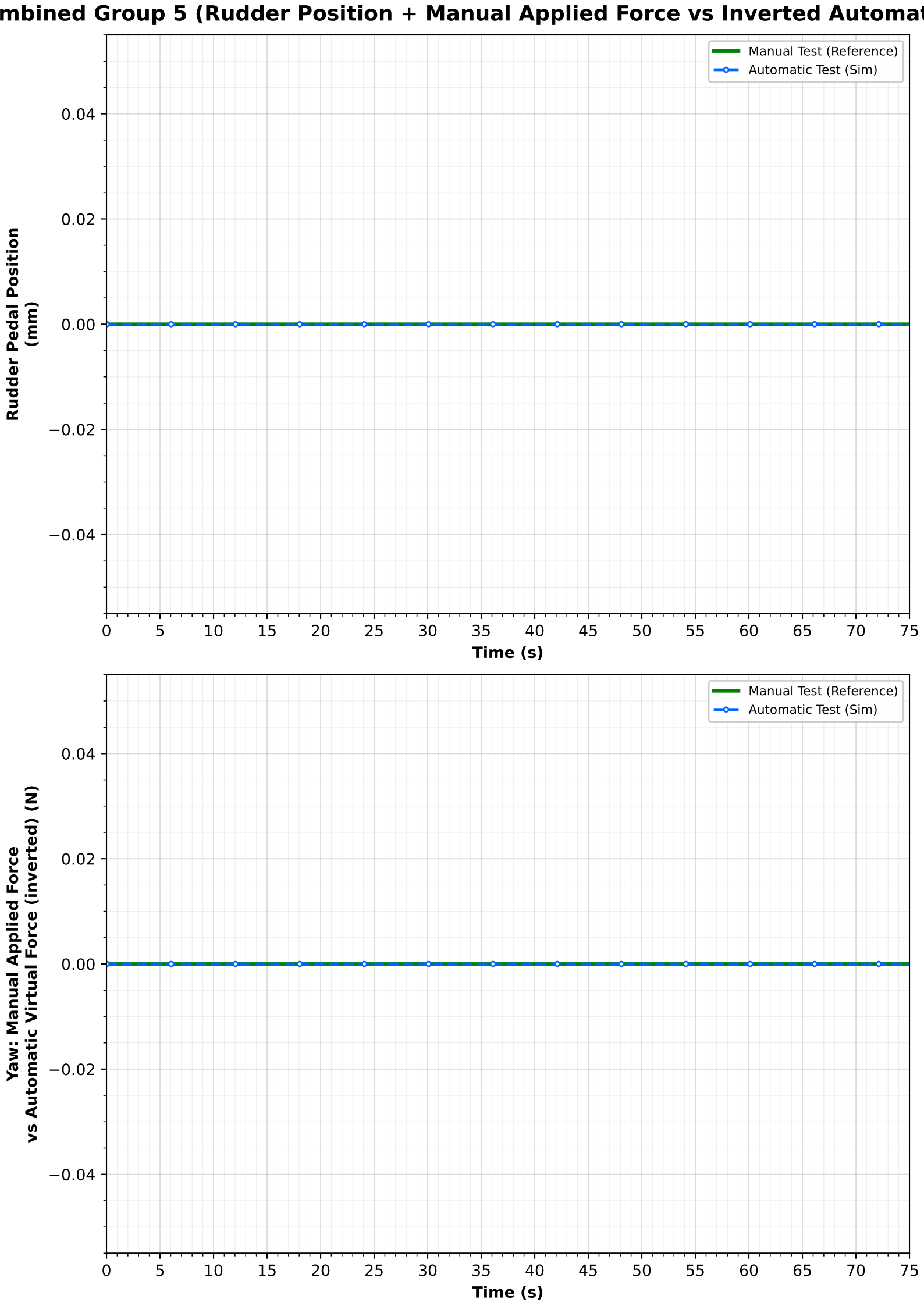
# 1st Quadrant Combined Group 3 (Propeller, Engine, Throttle)




t Combined Group 4 (Position + Manual Applied Force vs Inverted Automatic V
Manual Test (Reference)
Automatic Test (Sim)
Joystick Pitch Position (mm)
Time (s)
Manual Test (Reference)
Automatic Test (Sim)
Pitch: Manual Applied Force vs Automatic Virtual Force (inverted) (N)
Time (s)
Manual Test (Reference)
Automatic Test (Sim)
Joystick Roll Position (mm)
Time (s)
Manual Test (Reference)
Automatic Test (Sim)
Roll: Manual Applied Force vs Automatic Virtual Force (inverted) (N)
Time (s)


mbined Group 5 (Rudder Position + Manual Applied Force vs Inverted Automat
Manual Test (Reference)
Automatic Test (Sim)
Rudder Pedal Position (mm)
0.04
0.02
0.00
−0.02
−0.04
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
Time (s)
Manual Test (Reference)
Automatic Test (Sim)
Yaw: Manual Applied Force vs Automatic Virtual Force (inverted) (N)
0.04
0.02
0.00
−0.02
−0.04
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
Time (s)